

Optimization Requirements

Parameters:

Name	Discription	Type	Initial Value	Lower Limit	Upper Limit	Step
Dr_out	Outer diameter of Rotor	Calculated	Ds_in-2*air_gap	-	-	-
Dr_in	Inner diameter of Rotor	Variable	90mm	50mm	90mm	5mm
L_stk	Stack Length	Constant	134	-	-	-
Speed_rpm	Rotational speed of rotor in Rotations per minute	Calculated	$\frac{120 \cdot f_0}{PolesNum}$	-	-	-
Air_gap	Length of air gap between rotor and stator	Variable	1mm	0.5mm	1.5mm	0.1mm
Lamda	Lstack to D_ag ratio	Variable	0.9	0.8	1	0.1
D_ag	Diameter of middle of air gap	Calculated	Lstk/lamda	-	-	-
Ds_out	Outer Diameter of Stator	Constant	240mm	-	-	-
Ds_in	Inner Diameter of Stator	Calculated	D_ag + air_gap	-	-	-
Bridge	Distance between outer rotor diameter to magnet pole "holes"	Variable	1.5mm	1mm	3mm	0.1mm

SlotNum	Number of stator Slots	Constant	36	-	-	-
A_slot	Slot Area	Measured	213.1743016 mm ²	-	-	-
Hs0	Tooth slot height	Variable	1.1899mm	1mm	2mm	0.1mm
Hs1	Slot “slope”	Variable	1.5mm	1mm	2mm	0.1mm
Hs2	Slot height	Variable	18.07656mm	16mm	Hs0+Hs1+Hs2<[(Ds_out-Ds_in)/2]-12.25mm (38mm)	1mm
Bs0	Slot opening width	Variable	2.1128mm	1.5mm	4mm	0.5mm
Bs1	Lower Slot width	Variable	6.90142mm	3mm	10mm	0.5mm
Bs2	Upper Slot width	Variable	10.88076	5mm	14mm	1mm
PolesNum	Number of rotor poles	Constant	6	-	-	-
O1	Distance between duct bottom	Variable	5.4mm	0mm	13mm	1mm
O2	Duct dulance from iner rotor diameter	Variable	6mm	2mm	7mm	0.5mm
B1	Duct thicknes	Variable	3.5mm	3.2mm	Mt-0.3mm (5.7mm)	0.5mm
rib	Rib width	Variable	2mm	2mm	15mm	1mm
Hrib	Rib height	Variable	2.4mm	2mm	6mm	0.5mm
Mt	Magnet thickness	Variable	5.282mm	4mm	6mm	0.2mm
Mw	Magnet width	Variable	25.44156mm	10mm	30mm	2mm
magDmin	Minimum distance between magnets	Variable	10mm	0mm	10mm	1mm

Commented [SP1]: Approximation hs2*bs2 (underhoots)

Commented [SP2]: Decided based on min{Ds_in} so we don't lose valuable samples on optimization and when its too big its just not gonna run that model.

	(in same pole)					
D1	Minimum diameter of Permanent magnet ducts	calculate d	$Ds_{in} - 2 \cdot air_gap - 2 \cdot bridge$	-	-	-
Acond	Area of single conductor	Calculate d	I_{max}/I	-	-	-
N	Number of conductors	Calculate d	$\left\lceil 0.7 \cdot \frac{Aslor}{Acond} \right\rceil$	-	-	-
I _{max}	Input current maximum	constant	200A	-	-	-
J		constant	5.5A	-	-	-
f0	Current frequency	constant	50Hz	-	-	-
t0	Time when we achieve maximum torque	calculate d	0ms	-	-	-
Thet	Current excitation phase shift in radians	calculate d	$\frac{thet_deg \cdot \pi}{180}$	-	-	-
Thet_deg	Current excitation phase shift in degrees	Variable	30°	0°	90°	1°
-	Rotor initial angle	constant	-20°	-	-	-

Equations:

Efficiency:

$$PowerEfficiency = \frac{P_{out}}{P_{in}}$$

Unit: None

Goal: Maximize

Power density:

$$PowerDensity = \frac{P_{out}}{W_{total}}$$

Unit: W/Kg

Goal: Maximize

Material cost:

$$TotalCost = \sum V_i \cdot costPerVolume_i, \quad i \in \{all\ materials\ used\}$$

Unit: \$

Goal: Minimize

Torque ripples:

$$TorqueRipples = \frac{pk2pk(Torque)}{mean(Torque)} * 100$$

Unit: %

Goal: Minimize

Flux linkage:

Total airgap flux:

$$\varphi_{total} = B_{av} \pi D s_{out} L_{stk},$$

B_{av} : Magnetic Loading

Stator Tooth flux:

$$\varphi_{st} = \frac{\varphi_{total}}{SlotNum}$$

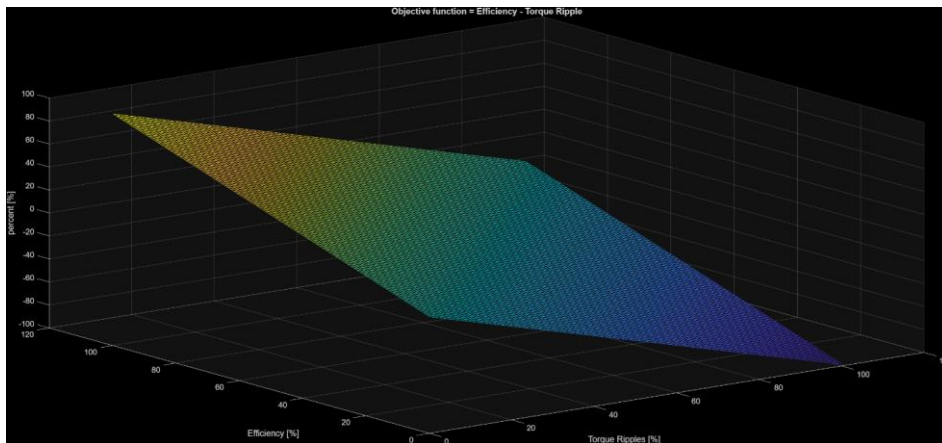
Pole flux:

$$\varphi_p = \frac{\varphi_{total}}{PolesNum}$$

Unit: Wb
Goal: maximize

Objective function:

Maximize Efficiency,
Minimize Torque ripple



By subtracting the torque ripple from the Efficiency, we get the above 3d mesh so we can see that if we use the objective function

$$\text{Objective function} = \text{Efficiency} - \text{Torque ripples}$$

The designs with high efficiency and low Torque ripples have a higher score than the ones with low efficiency and torque ripples.